Introduction to Mathematics and Physics

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Abstract

This is a simple abstract for the Introduction to Mathematics and Physics notes. It provides an overview of the topics covered in the course.

1 Lecture 1

2 Lecture 2

2.1 Introduction and the Geometric Picture

2.1.1 Basic definition

Definition 2.1

An ordinary differential equation (ODE) of order n is an equation of the form

$$F\left(x, y, y', \ldots, y^{(n)}\right) = 0,$$

where $y = y(x)$ is an unknown function and primes denote derivatives with respect to x.

Definition 2.2

A solution on an interval I is a function

$$y\colon I \to \mathbb{R}$$

that is n time differentiable and satisfies the differential equation for all $x \in I$.

Definition 2.3

The initial value problem (IVP) for an ordinary differential equation of order n is an equation of the form

$$F\left(x, y, y', \ldots, y^{(n)}\right) = 0,$$

together with initial conditions

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \ldots, \quad y^{(n-1)}(x_0) = y_{n-1},$$

where $x_0 \in I$ and $y_0, y_1, \ldots, y_{n-1} \in \mathbb{R}$.

Remark

For first-order ordinary differential equations

$$y' = f(x, y)$$

the initial value problem takes the form

$$y' = f(x, y), \quad y(x_0) = y_0$$

Example

Consider the first-order ordinary differential equation

$$y' = -y, \quad y(0) = 3.$$

The solution is

$$y = 3e^{-x}.$$

2.1.2 Direction Fields

The equation

$$y' = f(x, y)$$

shows that for each point (x, y) in the plane, the slope that any solution curve passing through that point must have, given by $f(x, y)$.

We can represent this information graphically using a direction field, where at each point (x, y) in the plane, we draw a small line segment with the slope $f(x, y)$.

Definition 2.4

A direction field (or slope field) for the first-order differential equation

$$y' = f(x, y)$$

is a graphical representation where at each point (x, y) in the plane, a small line segment is drawn with the slope $f(x, y)$.

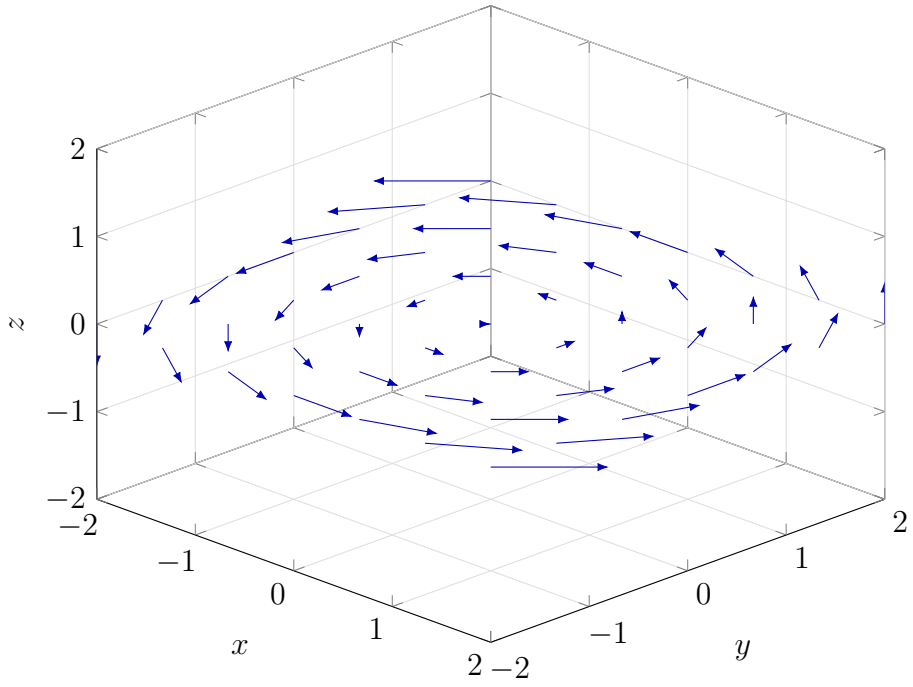
Solutions to the differential equation can be visualized as curves that are tangent everywhere to the line segments in the direction field.

Remark

Geometric Principle

Through each point (x, y) in the plane, there should be exactly one solution curve passing through it. Provided f is well-behaved (continuous and satisfies the Lipschitz condition) near the point (x, y). The Existence and Uniqueness Theorem make this principle precise.

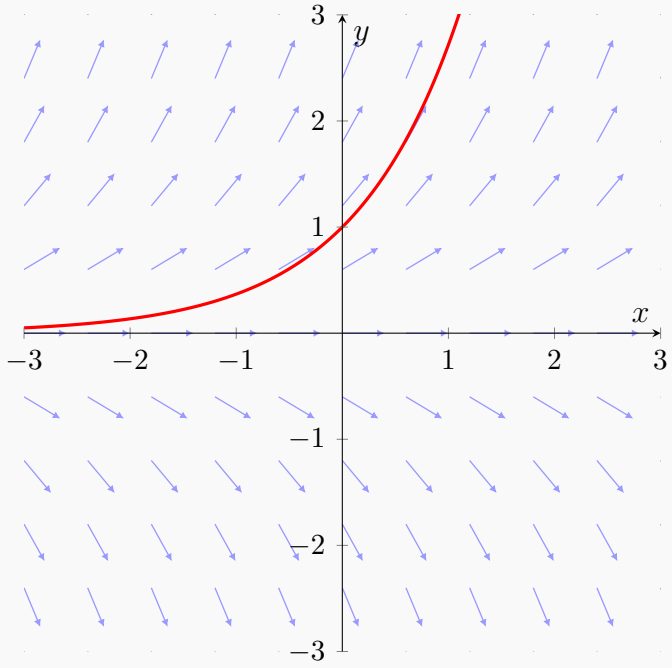
Vector field $\mathbf{F}(x, y, z) = (-y, x, 0)$



Example

For $y' = y$, the direction field consists of line segments with slope equal to the value of y at each point (x, y) in the plane.

Direction field of $y' = y$



2.2 Existence and Uniqueness

2.2.1 Motivation

Before spending time solving IVPs, we should ask whether a solution exists and whether it is unique.

Example

Consider the initial value problem

$$y' = y^{1/3}, \quad y(0) = 0.$$

The solution of this initial value problem is

$$y = \left(\frac{2}{3}x\right)^{3/2}.$$

And

$$y = 0.$$

Which is not unique.

2.2.2 The Picard-Lindelöf Theorem

Definition 2.5

Lipschitz Continuous

There exists a constant $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2| \quad \forall (x, y_1), (x, y_2) \in D.$$

Theorem 2.1

Picard-Lindelöf Theorem (Existence and Uniqueness)

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let $f(x, y)$ be defined on a rectangle D, suppose:

1. $f(x, y)$ is continuous on D.
2. $f(x, y)$ is Lipschitz continuous in y on D.

Then there exists $h > 0$ such that the initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0$$

has a unique solution $y(x)$ on the interval $|x - x_0| < h$.

Remark

A sufficient condition for the Lipschitz condition is that the partial derivative of f with respect to y exists and is bounded on D.

$$L = \sup_D \left| \frac{\partial f}{\partial y} \right|$$

2.2.3 proof scetch

Definition 2.6

sequence of functions defined by the Picard iteration:

$$y_0(x) = y_0, \quad y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))\, dt.$$

Each y_{n+1} is obtained by plugging the previous approximation into the right-hand side.

Key steps of the proof:

1. **Well-definedness.** If y_n stays in D, continuity of f ensures y_{n+1} is well defined and differentiable.

2. **Convergence.** Using the Lipschitz condition, one shows

$$|y_{n+1}(x) - y_n(x)| \leq \frac{ML^n |x - x_0|^{n+1}}{(n+1)!},$$

where $M = \sup_D |f|$. This is the n-th term of the series for $Me^{L|x-x_0|}/L$, so the series $(y_{n+1} - y_n)$ converges absolutely and uniformly on $|x - x_0| \leq h$ for h small enough.

3. **Limit is a solution.** Uniform convergence justifies passing the limit inside the integral; the limit function satisfies the integral equation, hence the IVP.

4. **Uniqueness.** If y and z are two solutions, then $u = y - z$ satisfies

$$|u(x)| \leq L \int |u(t)|\, dt.$$

Gronwall's inequality then forces $u \equiv 0$.

Example

(Picard iteration for $y' = y$, $y(0) = 1$)

$$
\begin{aligned}
y_0(x) &= 1, \\
y_1(x) &= 1 + \int_0^x y_0(t)\,dt = 1 + \int_0^x 1\,dt = 1 + x, \\
y_2(x) &= 1 + \int_0^x y_1(t)\,dt = 1 + \int_0^x (1 + t)\,dt = 1 + x + \frac{x^2}{2}, \\
y_3(x) &= 1 + \int_0^x y_2(t)\,dt = 1 + \int_0^x \left(1 + t + \frac{t^2}{2}\right) dt = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}, \\
&\vdots \\
y_n(x) &\to e^x \text{ as } n \to \infty.
\end{aligned}
$$

The partial sums if e^x emerge. The sequence converges to $y(x) = e^x$.

2.3 Separable Equations

Definition 2.7

A first-order ODE is called **separable** if $f(x, y)$ can be factored as a product of a function of x and a function of y.

So it can be written in the form

$$\frac{dy}{dx} = g(x)h(y),$$

where g and h are continuous functions.

The main idea of solving separable equations is separating the variables and integrating both sides.

1. Write the equation as $\dfrac{dy}{h(y)} = g(x)\,dx$ (formally separate variables).
2. Integrate both sides: $\displaystyle\int \frac{dy}{h(y)} = \int g(x)\,dx + C$.
3. Solve for y if possible (explicit solution); otherwise leave implicit.
4. Apply any initial condition to determine C.
5. Check for *equilibrium solutions* $h(y) = 0$ that may have been lost when dividing.

Remark

When we divide by $h(y)$, we implicitly assume $h(y) \neq 0$. Any constant value $y = c$ where $h(c) = 0$ is a solution (an *equilibrium solution*) and must be checked separately.

2.4 Substitution method

2.4.1 Homogeneous Equations

Definition 2.8

A first-order ODE is called **homogeneous** if it can be written in the form

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

For some function f.

The substitution $v = \frac{y}{x}$ (hence $y = vx$) can be used to transform it into a separable equation.

1. Verify the equation has the form $y' = F(y/x)$.
2. Substitute $v = y/x$, so $y = vx$ and $y' = v + xv'$.
3. The equation becomes $v + xv' = F(v)$, i.e. $x\frac{\mathrm{d}v}{\mathrm{d}x} = F(v) - v$, which is **separable** in v and x.
4. Solve for $v(x)$, then back-substitute $y = vx$.

2.4.2 Bernoulli Equations

Definition 2.9

A first-order ODE is called a **Bernoulli equation** if it can be written in the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n,$$

where $n \neq 0, 1$ and P and Q are continuous functions. (If $n = 0$ or $n = 1$, the equation is linear or separable, respectively.)

The substitution $v = y^{1-n}$ (hence $y = v^{\frac{1}{1-n}}$) can be used to transform it into a linear equation.

1. Divide through by y^n: $\quad y^{-n}y' + P(x)y^{1-n} = Q(x)$.
2. Substitute $w = y^{1-n}$, so $w' = (1-n)y^{-n}y'$.
3. The equation becomes the **first-order linear** equation

$$\frac{w'}{1-n} + P(x)w = Q(x), \quad \text{i.e.} \quad w' + (1-n)P(x)w = (1-n)Q(x).$$

4. Solve for $w(x)$, then recover $y = w^{1/(1-n)}$.

3 Lecture 3

3.1 First Order Linear Equations

Definition 3.1

A first-order linear equation is an ODE of the form

$$\frac{dy}{dx} + P(x)y = Q(x),$$

where P and Q are continuous functions.

- The equation is called homogeneous if $Q \equiv 0$.
- The equation is called non-homogeneous (or inhomogeneous) otherwise.

Remark

In this context, "homogeneous" means $|Q| = 0$, not the "F (y/x)" sense.

3.1.1 Homogeneous Equations

The homogeneous first-order linear equation is of the form

$$\frac{dy}{dx} + P(x)y = 0.$$

Which is separable.

$$\begin{aligned}
\frac{dy}{dx} &= -P(x)y, \\
\frac{dy}{y} &= -P(x)dx. \\
\int \frac{dy}{y} &= \int -P(x)dx. \\
\ln|y| &= -\int P(x)dx + C, \\
y &= Ce^{-\int P(x)dx}.
\end{aligned}$$

where C is an arbitrary constant. This is the complementary solution (or homogeneous solution) y_h.

Theorem 3.1

Superposition principle

Structure of the general solution of a first-order linear equation If y_p is a particular solution of the non-homogeneous equation

$$\frac{dy}{dx} + P(x)y = Q(x),$$

then the general solution is

$$y = y_h + y_p,$$

where y_h is the complementary solution of the corresponding homogeneous equation.

Verify this by substituting $y = y_h + y_p$ into the original non-homogeneous equation.

$$\begin{aligned}
\frac{d}{dx}(y_h + y_p) + P(x)(y_h + y_p) &= \frac{dy_h}{dx} + \frac{dy_p}{dx} + P(x)y_h + P(x)y_p \\
&= \underbrace{\frac{dy_h}{dx} + P(x)y_h}_{=0} + \frac{dy_p}{dx} + P(x)y_p \\
&= \frac{dy_p}{dx} + P(x)y_p \\
&= Q(x).
\end{aligned}$$

The superposition principle states that the general solution of a non-homogeneous first-order linear equation is the sum of the complementary solution and a particular solution. It is a fundamental feature of all linear equations and has no analogue for nonlinear ones.

3.2 The Method of Integrating Factor

3.2.1 Definition

We want μ such that

$$\mu(x)[y' + P(x)y] = [\mu(x)y]'.$$

Expanding the right side

$$[\mu y]' = \mu' y + \mu y'$$

Matching with the left side

$$\mu y' + \mu P y = \mu' y + \mu y' \implies \mu P = \mu',$$

so

$$\frac{\mu'}{\mu} = P(x) \implies \mu(x) = e^{\int P(x)\,dx}$$

Definition 3.2

The function

$$\mu(x) = e^{\int P(x)\,dx}$$

is the integrating factor for the first-order linear differential equation

$$y' + P(x)y = Q(x)$$

The general solution of the differential equation can be expressed as

$$y = \frac{1}{\mu(x)}\left(\int \mu(x)Q(x)\,dx + C\right),$$

where C is an arbitrary constant.

Proof. We start with the first-order linear differential equation

$$y' + P(x)y = Q(x).$$

Then multiply both sides of the differential equation by $\mu(x)$

$$\mu(x)[y' + P(x)y] = [\mu(x)y]'.$$

the general solution can then be found by integrating both sides with respect to x.

$$\begin{aligned}
&[\mu(x)y]' = \mu(x)Q(x) \\
&\mu(x)y = \int \mu(x)Q(x)\,dx + C \implies y = \frac{1}{\mu(x)}\left(\int \mu(x)Q(x)\,dx + C\right),
\end{aligned}$$

where C is an arbitrary constant.

$$y = \frac{1}{\mu(x)}\left(\int \mu(x)Q(x)\,dx + C\right)$$

□

3.2.2 Example

Example

Solve the first-order linear differential equation $y' + 2y = 4$, $y(0) = 1$.

$$P(x) = 2, \quad Q(x) = 4$$

The integrating factor is

$$\mu(x) = e^{\int 2\,dx} = e^{2x}.$$

Multiply both sides by $\mu(x)$

$$e^{2x}y' + 2e^{2x}y = 4e^{2x} \implies (e^{2x}y)' = 4e^{2x}.$$

Integrate both sides

$$e^{2x}y = \int 4e^{2x}\,dx + C = 2e^{2x} + C \implies y = 2 + Ce^{-2x}.$$

Applying the initial condition $y(0) = 1$,

$$y = 2 - e^{-2x}.$$

Example

Solve the first-order linear differential equation $y' + 2xy = x^2$, $y(1) = 0$ for $x > 0$.

$$P(x) = 2x, \quad Q(x) = x^2$$

The integrating factor is

$$\mu(x) = e^{\int 2x\,dx} = e^{x^2}.$$

Multiply both sides by $\mu(x)$

$$e^{x^2}y' + 2xe^{x^2}y = x^2e^{x^2} \implies (e^{x^2}y)' = x^2e^{x^2}.$$

Integrate both sides

$$e^{x^2}y = \int x^2e^{x^2}\,dx + C \implies y = e^{-x^2}\left(\int x^2e^{x^2}\,dx + C\right).$$

Applying the initial condition $y(1) = 0$,

$$0 = e^{-1}\left(\int_1^1 x^2e^{x^2}\,dx + C\right) \implies C = 0.$$

Therefore, the solution is

$$y = e^{-x^2}\int_1^x t^2e^{t^2}\,dt.$$

Remark

Algorithm for solving the first-order linear differential equation $y' + P(x)y = Q(x)$

1. Identify $P(x)$ and $Q(x)$ from the differential equation $y' + P(x)y = Q(x)$.
2. Compute the integrating factor $\mu(x) = e^{\int P(x)\,dx}$.
3. Multiply both sides by $\mu(x)$ to obtain $[\mu(x)y]' = \mu(x)Q(x)$.
4. Integrate both sides with respect to x

$$\mu(x)y = \int \mu(x)Q(x)\,dx + C$$

.

5. Solve for y by dividing both sides by $\mu(x)$

$$y = \frac{1}{\mu(x)}\left(\int \mu(x)Q(x)\,dx + C\right)$$

.

6. Apply initial conditions if given to determine the constant C.

3.2.3 Variation of Parameters

There is another approach to solving first-order linear differential equations called the method of variation of parameters.

Definition 3.3

The method of variation of parameters is a technique for finding a particular solution to a nonhomogeneous first-order linear differential equation of the form

$$y' + P(x)y = Q(x)$$

The general solution is given by

$$y = y_h + y_p,$$

where y_h is the solution to the corresponding homogeneous equation $y' + P(x)y = 0$, and y_p is a particular solution found by assuming $y_p = u(x)y_h$ and determining $u(x)$ such that y_p satisfies the original nonhomogeneous equation.

We know the homogeneous solution is

$$y_h = Ae^{-\int P(x)\,dx}.$$

The idea: treat A as a functionof x and seek a particular solution $y_p = A(x)e^{-\int P(x)\,dx}$. Computing y_p' and substituting into

$$y' + P(x)y = Q(x)$$

we obtain an equation for $A'(x)$

$$A'(x)e^{-\int P(x)\,dx} = Q(x) \implies A'(x) = Q(x)e^{\int P(x)\,dx}.$$

Integrating both sides gives

$$A(x) = \int Q(x)e^{\int P(x)\,dx}\,dx.$$

Therefore, the particular solution is

$$y_p = A(x)e^{-\int P(x)\,dx} = e^{-\int P(x)\,dx}\int Q(x)e^{\int P(x)\,dx}\,dx.$$

3.3 Existence and Uniqueness of Differential Equations

3.3.1 The Sharper Theorem

Theorem 3.2

Suppose $P(x)$ and $Q(x)$ are continuous on an open interval I containing x_0. Then the IVP

$$y' + P(x)y = Q(x), \quad y(x_0) = y_0$$

has a unique solution $y(x)$ that exists for all $x \in I$.

Proof. The proof of this theorem relies on the method of integrating factors. Consider the differential equation

$$y' + P(x)y = Q(x), \quad y(x_0) = y_0.$$

The integrating factor

$$\mu(x) = e^{\int P(x)\,dx}$$

is well-defined and continuously differentiable on I (since P is continuous).
Then $\mu(x)Q(x)$ is continuous on I, so

$$H(x) = \int_{x_0}^{x} \mu(t)Q(t)\,dt$$

is a well-defined C^1 function on I.

$$\frac{d}{dx}\left(\mu(x)y\right) = \mu(x)Q(x)$$

Is Differentiable, and integrating both sides gives

$$\mu(x)y = \int \mu(x)Q(x)\,dx + C$$

$$y = \frac{1}{\mu(x)}\left(\int \mu(x)Q(x)\,dx + C\right).$$

And I needs to contain the initial condition $y(x_0) = y_0$ to determine the constant C.
Specifically, substituting $x = x_0$ and $y = y_0$ into the solution gives

$$y_0 = \frac{1}{\mu(x_0)}\left(\int_{x_0}^{x_0} \mu(t)Q(t)\,dt + C\right) = \frac{C}{\mu(x_0)} \implies C = y_0\mu(x_0).$$

Therefore, the unique solution satisfying the initial condition is

$$y = \frac{1}{\mu(x)}\left(\int_{x_0}^{x} \mu(t)Q(t)\,dt + y_0\mu(x_0)\right).$$

□

3.3.2 Global Existence

Definition 3.4

A solution $y(x)$ of the IVP

$$y' + P(x)y = Q(x), \quad y(x_0) = y_0$$

is said to have *global existence* if it exists for all $x \in \mathbb{R}$.

Remark

Why global existence holds for linear equations but not necessarily for nonlinear ones.

Contrast with the nonlinear case. For the nonlinear IVP

$$y' = y^2, y(0) = 1$$

the solution is

$$y = \frac{1}{1-x}$$

which blows up at $x = 1$, even though $f(x, y) = y^2$ is perfectly smooth everywhere. Solutions of nonlinear equations can escape to infinity in finite time, even when f is analytic.
For the linear case

$$y' + P(x)y = Q(x)$$

with P, Q continuous on $\mathbb{R}$, the solution exists for all $x \in \mathbb{R}$. Global existence is guaranteed.

3.3.3 The General Solution is Complete

Definition 3.5

The general solution of a linear first-order differential equation

$$y' + P(x)y = Q(x)$$

is complete in the sense that it contains all possible solutions. Any solution can be obtained by choosing an appropriate constant C in the general solution

$$y = \frac{1}{\mu(x)} \left(\int_{x_0}^{x} \mu(t)Q(t)\, dt + C \right),$$

where $\mu(x) = e^{\int P(x)\, dx}$ is the integrating factor.

3.4 Applications

3.4.1 R-C Circuit

Consider an R-C circuit with a resistor R and a capacitor C in series, connected to a voltage source $V(t)$. The governing equation for the charge $q(t)$ on the capacitor is

$$R\frac{dq}{dt} + \frac{q}{C} = V(t).$$

This is a linear first-order differential equation of the form

$$q' + \frac{1}{RC}q = \frac{V(t)}{R}.$$

This is a first-order linear ODE with $P = 1/(RC)$ and $Q = V(t)/R$.

3.4.2 Mixing Problem

Consider a tank initially containing V_0 liters of brine with a salt concentration of C_0 grams per liter. Brine with a concentration of C_{in} grams per liter flows into the tank at a rate of r_{in} liters per minute, and the well-mixed solution flows out at a rate of r_{out} liters per minute. Let $C(t)$ denote the concentration of salt in the tank at time t. The governing equation is

$$V_0\frac{dC}{dt} = r_{\text{in}}C_{\text{in}} - r_{\text{out}}C.$$

This is a linear first-order differential equation of the form

$$C' + \frac{r_{\text{out}}}{V_0}C = \frac{r_{\text{in}}C_{\text{in}}}{V_0}.$$

This is a first-order linear ODE with $P = r_{\text{out}}/V_0$ and $Q = r_{\text{in}}C_{\text{in}}/V_0$.

3.5 Summary

3.5.1 Topic

Topic	**Key condition**	**Result**
Picard-Lindelöf	f continuous + Lipschitz	Local existence & uniqueness
Separable equations	$y' = g(x)h(y)$	Separate, integrate, check equilibria
Homogeneous	$y' = F(y/x)$	Substitute $v = y/x$, separable
Bernoulli	$y' + Py = Qy^n$	Substitute $w = y^{1-n}$, linearise
Linear (integrating factor)	P, Q continuous on I	$\mu = e^{\int P dx}$, global solution
Linear existence & uniqueness	P, Q continuous on I	Unique solution on all of I

3.5.2 Picture

ODEs can be broadly classified into linear and nonlinear types.

On the general side, we have local existence and uniqueness under the Picard-Lindelöf conditions, It have such Complex behaviour, including possible finite-time blow-up and non-uniqueness.

On the linear side, we have explicit global solutions using the integrating factor method and a sharp uniqueness theorem. Where the P and Q are continuous, the solution exists and is unique on the entire interval.

The linear case is much simpler case and easier to handle analytically.

4 Lecture 4: Functions of Several Variables

Example

Temperature in this room

$$T = T(x, y, z) \quad \text{temperature field}$$

a three-variable function of x, y, z.

4.1 Function of Several Variables

4.1.1 Scalar fields

Definition 4.1

Let $D \subseteq \mathbb{R}^n$. A real function of n variables

$$f \colon D \to \mathbb{R}$$
$$(x_1, x_2, \ldots, x_n) \mapsto z = f(x_1, x_2, \ldots, x_n) \in \mathbb{R}$$

is called a *scalar field*.

Remark

It is a linear functional of its arguments if it is linear.

Example

For the 2-dimensional case, a scalar field is a function

$$f \colon D \subseteq \mathbb{R}^2 \to \mathbb{R}$$
$$(x, y) \mapsto f(x, y)$$

4.1.2 Level sets

Definition 4.2

The *level set* at value c of f is the set

$$\{(x_1, x_2, \ldots, x_n) \in D \mid f(x_1, x_2, \ldots, x_n) = c\}.$$

Example

Consider the potential field

$$V(r) = \frac{1}{r}.$$

Then

$$\begin{aligned} V(x_1, x_2, x_3) &= \frac{1}{\sqrt{x_1^2 + x_2^2 + x_3^2}} \\ &= \left(\sum_{i=1}^{3} x_i^2 \right)^{-\frac{1}{2}}. \end{aligned}$$

The level sets of this potential field satisfy

$$V(x_1, x_2, x_3) = c \implies \left(\sum_{i=1}^{3} x_i^2 \right)^{\frac{1}{2}} = c^{-1}.$$

Three variables and one equation give just one degree of freedom, which corresponds to a 2-dimensional surface in $\mathbb{R}^3$.

4.2 Partial Derivatives

4.2.1 First order partial derivatives

Definition 4.3

The partial derivative of $f(x, y)$ with respect to x is defined as

$$\begin{aligned} \frac{\partial f}{\partial x} = \lim_{\epsilon \to 0} \frac{f(x + \epsilon, y) - f(x, y)}{\epsilon} &= f_x(x, y) \\ &= \frac{\partial f}{\partial x}(x, y) = \left. \frac{\partial f}{\partial x} \right|_{(x,y)} \end{aligned}$$

if the limit exists.

$$\iff f(x + \epsilon, y) = f(x, y) + \epsilon f_x(x, y) + \mathcal{O}(\epsilon^2).$$

The partial derivative of $f(x, y)$ with respect to y is defined similarly:

$$\frac{\partial f}{\partial y} = \lim_{\epsilon \to 0} \frac{f(x, y + \epsilon) - f(x, y)}{\epsilon} = f_y(x, y) = \frac{\partial f}{\partial y}(x, y) = \left. \frac{\partial f}{\partial y} \right|_{(x,y)}.$$

Proof.

$$\begin{aligned}\lim_{\epsilon\to 0}\frac{f(x,y)+\epsilon f_x(x,y)+\mathcal{O}(\epsilon^2)-f(x,y)}{\epsilon} &= \lim_{\epsilon\to 0}\frac{\epsilon f_x(x,y)+\mathcal{O}(\epsilon^2)}{\epsilon}\\ &= \lim_{\epsilon\to 0}\left(f_x(x,y)+\mathcal{O}(\epsilon)\right)\\ &= f_x(x,y).\end{aligned}$$

□

Example

Let

$$f(x,y) = x^3y + \sin(xy).$$

Then

$$\begin{aligned}\frac{\partial f}{\partial x} &= 3x^2y + y\cos(xy),\\ \frac{\partial f}{\partial y} &= x^3 + x\cos(xy).\end{aligned}$$

4.2.2 Higher order partial derivatives

Definition 4.4

The second order partial derivatives of $f(x,y)$ are

$$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \quad f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right), \quad f_{xy} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right), \quad f_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

Theorem 4.1

Schwarz's theorem

If f_{xy} and f_{yx} exist and are continuous on D, then

$$f_{xy}(x,y) = f_{yx}(x,y) \qquad \forall (x,y) \in D.$$

4.3 Gradient

4.3.1 Gradient

Definition 4.5

The gradient of f is the vector of its first-order partial derivatives:

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) \in \mathbb{R}^n,$$

an n-dimensional vector.

We have

$$\begin{aligned} f(x+\Delta x, y+\Delta y) &= f(x, y+\Delta y) + f_x(x, y+\Delta y)\Delta x + \mathcal{O}(\Delta x) \\ &= f(x,y) + f_y(x,y)\Delta y + \mathcal{O}(\Delta y^2) \\ &\quad + f_x(x,y)\Delta x + f_{xy}(x,y)\Delta x \Delta y + \mathcal{O}(\Delta x^2) \\ &= f(x,y) + [f_x(x,y)\Delta x + f_y(x,y)\Delta y] + \mathcal{O}(\epsilon^2), \end{aligned}$$

where $\Delta x, \Delta y \sim \epsilon$.
More generally,

$$\begin{aligned} f(x_1+\Delta x_1, \ldots, x_n+\Delta x_n) - f(x_1, \ldots, x_n) &= \sum_{i=1}^{n} \left.\frac{\partial f}{\partial x_i}\right|_{(x_1,\ldots,x_n)} \Delta x_i + \mathcal{O}(\epsilon^2) \\ &= \Delta f(x_1, \ldots, x_n). \end{aligned}$$

Let $\vec{x} = (x_1, \ldots, x_n)$.

4.3.2 Total Differential

Definition 4.6

The total differential of f is

$$df = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}\, dx_j,$$

where $dx_i = \Delta x_i$. Thus

$$\begin{aligned} df &= \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}\, dx_j \\ &= \nabla f \cdot d\vec{x}. \end{aligned}$$

Remark

The total differential of a function f at a point $\vec{x} = (x_1, \ldots, x_n)$ is the linear approximation of the change in f given by

$$df = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} dx_j$$

4.3.3 Infinitesimal variation in direction $\vec{u}$

Definition 4.7

The directional derivative of f in the direction of a unit vector $\vec{u}$ is defined as

$$D_{\vec{u}} f(\vec{x}) = \lim_{\epsilon \to 0} \frac{f(\vec{x} + \epsilon\vec{u}) - f(\vec{x})}{\epsilon}$$

Which is

$$D_{\vec{u}} f(\vec{x}) = \nabla f \cdot \vec{u}.$$

Assume $\|\vec{u}\| = 1$. Then

$$\Delta f = f(\vec{x} + \epsilon\vec{u}) - f(\vec{x}).$$

This is no different from hiking a mountain along the direction of $\vec{u}$.

$$\begin{aligned}\Delta f &= f(\vec{x}+\epsilon\vec{u}) - f(\vec{x}) \\ &= \epsilon\left[\frac{\partial f}{\partial x_1}u_1 + \cdots + \frac{\partial f}{\partial x_n}u_n\right] + \mathcal{O}(\epsilon^2) \\ &= \epsilon\nabla f \cdot \vec{u} + \mathcal{O}(\epsilon^2).\end{aligned}$$

The directional derivative is

$$D_{\vec{u}}f(\vec{x}) = \lim_{\epsilon\to 0}\frac{f(\vec{x}+\epsilon\vec{u}) - f(\vec{x})}{\epsilon} = \nabla f \cdot \vec{u} \quad \text{(a scalar)}.$$

Properties

where θ is the angle between ∇f and $\vec{u}$.

$$\nabla f \cdot \vec{u} = \|\nabla f\| \cos\theta,$$

So the gradient points in the direction of the steepest ascent of the function.

1. Maximum value when $\vec{u}$ is along ∇f.
2. ∇f is orthogonal to the tangent to the level set.

4.3.4 The Chain Rule

Definition 4.8

For $n = 2$, let $f(x, y)$, $x(t)$, $y(t)$, and

$$F(t) = f(x(t), y(t))$$

Then

$$\frac{dF}{dt} = \lim_{\epsilon\to 0}\frac{F(t+\epsilon) - F(t)}{\epsilon} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Generalize to n
let

$$f(x_1, \ldots, x_n), \quad x_i = x_i(t)$$

Then

$$\frac{df}{dt} = \sum_{j=1}^{n}\frac{\partial f}{\partial x_j}\frac{dx_j}{dt} = \nabla f \cdot \dot{\vec{x}}(t)$$

where $\dot{\vec{x}}(t) = \frac{d\vec{x}}{dt}$.

For $n = 2$, let

$$\begin{aligned}x &= x(s,t) \\ y &= y(s,t)\end{aligned}$$

and define

$$F(s,t) = f(x(s,t), y(s,t))$$

This transforms points from the (x, y)-plane to the (s, t)-plane.

Notation

In physics notation,

$$\frac{\partial F}{\partial s} = \text{“}\frac{\partial f}{\partial s}\text{”}.$$

We have

$$\begin{cases} \dfrac{\partial F}{\partial s} = \dfrac{\partial f}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial f}{\partial y}\dfrac{\partial y}{\partial s} \\ \dfrac{\partial F}{\partial t} = \dfrac{\partial f}{\partial x}\dfrac{\partial x}{\partial t} + \dfrac{\partial f}{\partial y}\dfrac{\partial y}{\partial t} \end{cases}$$

Generalize to n: let $(x_1, \dots, x_n) \to (\xi_1, \dots, \xi_n)$, with

$$\begin{cases} x_1 = x_1(\xi_1, \dots, \xi_n) \\ x_2 = x_2(\xi_1, \dots, \xi_n) \\ \quad \vdots \\ x_n = x_n(\xi_1, \dots, \xi_n) \end{cases}$$

Then

$$\frac{\partial f}{\partial \xi_i} = \sum_{k=1}^{n} \frac{\partial f}{\partial x_k}\frac{\partial x_k}{\partial \xi_i} = \sum_{k=1}^{n} (\nabla f)_k J_{ik}$$

$$\underbrace{\frac{\partial f}{\partial x_k}}_{\substack{\text{component of } \nabla f \\ \text{n-dim vector}}} \qquad \underbrace{\frac{\partial x_k}{\partial \xi_i}}_{\substack{\text{component of } J \\ \text{n-dim vector}}}$$

and

$$\frac{\partial f}{\partial x_i} = \sum_{k=1}^{n} (J^{-1})_{ik}\frac{\partial f}{\partial \xi_k}$$

Definition 4.9

The Jacobian matrix

$$J_{ik} = \frac{\partial x_k}{\partial \xi_i}$$

i.e.

$$J = \begin{pmatrix} \frac{\partial x_1}{\partial \xi_1} & \cdots & \frac{\partial x_n}{\partial \xi_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_1}{\partial \xi_n} & \cdots & \frac{\partial x_n}{\partial \xi_n} \end{pmatrix}_{n \times n}$$

Matrix-vector relations can be written as

$$\vec{v} = M \cdot \vec{w}, \qquad \vec{w} = M^{-1} \cdot \vec{v}$$

Example

Take (x_1, x_2), and we can also use (r, θ) to describe it, where

$$\begin{cases} x_1 = r\cos\theta \\ x_2 = r\sin\theta \end{cases}$$

Then

$$J = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} = \begin{pmatrix} \frac{\partial x_1}{\partial r} & \frac{\partial x_1}{\partial \theta} \\ \frac{\partial x_2}{\partial r} & \frac{\partial x_2}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

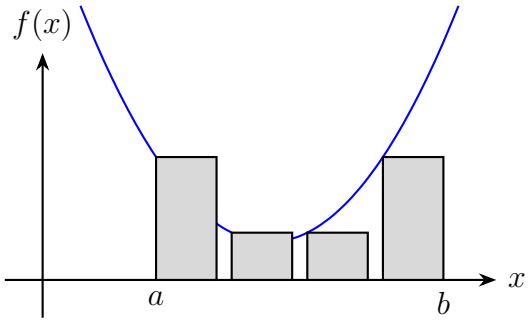
5 Lecture 5: Several Variable Integration and Linear Algebra

5.1 Multiple Integrals

5.1.1 One dimensional integrals

$$\int_a^b dx\, f(x)$$

Area under the curve is defined by the Riemann integral, can be calculated by the sum the small rectangles as the limit when their width approaches zero.



Definition 5.1

Riemann integral is defined as

$$\int_a^b f(x)\, dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x_i$$

is the real area under the curve $y = f(x)$ from $x = a$ to $x = b$.

5.1.2 Double Integrals

$$\int_A dxdy\, f(x, y)$$

Volume under the surface is defined by this double integral.

Definition 5.2

For

$$z = f(x, y),$$

Double integral is defined as

$$\int_A dxdy\, f(x,y) = \iint dxdy\, f(x,y) = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta A_{ij}$$

where A is the region in the xy-plane, and ΔA_{ij} is the area of the small subrectangle.

If the surface A is a rectangle

$$A = [a, b] \times [c, d],$$

then the double integral can be written as an iterated integral

$$\int_A dxdy\, f(x,y) = \int_a^b dy \underbrace{\left[\int_c^d dx\, f(x,y)\right]}_{F(y,a,b)}.$$

5.2 Change of variables

5.2.1 One Dimensional Change of Variables

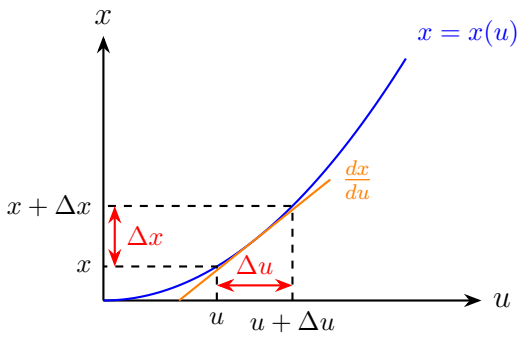
One dimensional change of variables is given by

$$x = x(u), \quad dx = \frac{dx}{du}du,$$

then

$$\int_a^b dx\, f(x) = \int_{u(a)}^{u(b)} du \frac{dx}{du} f(x(u)).$$

Local tilted



5.2.2 Two Dimensional Change of Variables

So for a linear transformation (change of variables)

$$T\colon R^2 \to \mathbb{R}^2$$
$$(u, v) \mapsto (x(u, v), y(u, v)).$$

Definition 5.3

The Jacobian of the transformation T is defined as

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

which simplifies to

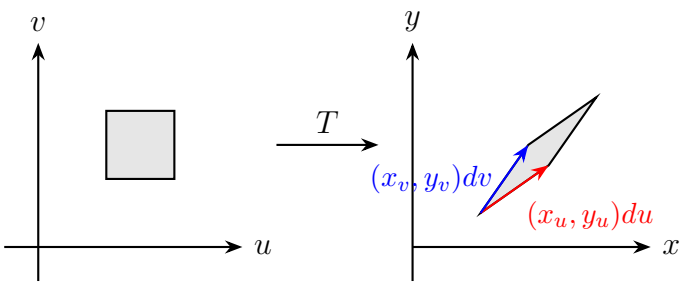
$$J = x_u y_v - x_v y_u.$$

Then the double integral under the change of variables is given by

$$\int_A dxdy\, f(x,y) = \int_S dudv\ |J|\, f(x(u,v), y(u,v)).$$

where

$$S = T^{-1}(A).$$

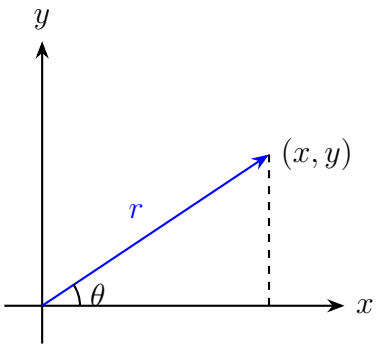


5.2.3 Polar coordinates

Definition 5.4

Polar coordinates are given by

$$x = r\cos\theta, \quad y = r\sin\theta.$$



The Jacobian is

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix}.$$

The determinant of this matrix is

$$|J| = \left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = r.$$

Therefore, the double integral in polar coordinates is

$$\int_A dxdy\, f(x,y) = \int_S drd\theta\, r\, f(r\cos\theta, r\sin\theta).$$

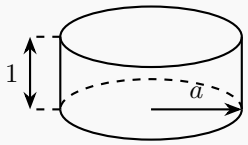
where $S = T^{-1}(A)$.

Example

Area of a circle of radius a

$$A = \pi a^2.$$

When prove it, we need to calculate the volumn of a cylinder of height 1.



$$\begin{aligned} V &= A \cdot 1 \\ &= \int_A dxdy\, 1 = \int_0^{2\pi} d\theta \int_0^a r\, dr = \int_0^{2\pi} d\theta\, \frac{a^2}{2} = \pi a^2. \end{aligned}$$

Example

Gaussian integral

$$I = \int_{-\infty}^{\infty} dx e^{-x^2} = \sqrt{\pi}.$$

The simplest way to evaluate this integral is to consider the square of the integral and switch to polar coordinates

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} dx e^{-x^2}\right)^2 \\ &= \int_{-\infty}^{\infty} dx\, e^{-x^2} \int_{-\infty}^{\infty} dy\, e^{-y^2} \\ &= \int_{\mathbb{R}^2} dxdy\, e^{-(x^2+y^2)}. \end{aligned}$$

Switching to polar coordinates, we have

$$I^2 = \int_0^{2\pi} d\theta \int_0^{\infty} r\, dr\, e^{-r^2} = 2\pi \int_0^{\infty} r\, dr\, e^{-r^2}.$$

Let $t = r^2$, then $dt = 2r\, dr$, so

$$I^2 = \pi \int_0^{\infty} dt\, e^{-t} = \pi.$$

Therefore,

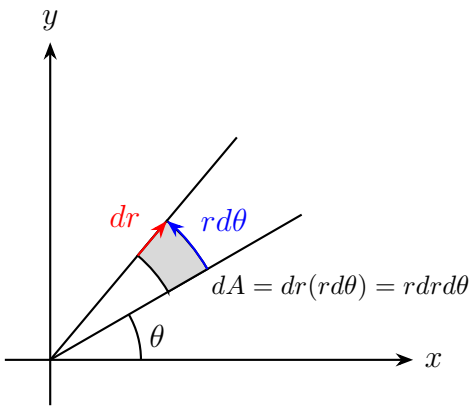
$$I = \sqrt{\pi}.$$

5.2.4 Inverse Transformation of Polar Coordinates

Definition 5.5

The inverse transformation of polar coordinates is given by

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan \frac{y}{x}.$$



Compute the Jacobian of the inverse transformation

$$\frac{\partial(r,\theta)}{\partial(x,y)} = \begin{pmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}$$

Take the determinant to find the Jacobian

$$\left|\frac{\partial(r,\theta)}{\partial(x,y)}\right| = \det \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix} = \frac{x^2+y^2}{r^3} = \frac{r^2}{r^3} = \frac{1}{r}.$$

This matrix is the inverse of the Jacobian matrix of the polar coordinate transformation (need to verify),

$$\frac{\partial(x,y)}{\partial(r,\theta)} \frac{\partial(r,\theta)}{\partial(x,y)} = I.$$

And

$$\frac{\partial \theta}{\partial y} \neq \left(\frac{\partial y}{\partial \theta}\right)^{-1}.$$

So

$$\frac{dx}{du} = \left(\frac{du}{dx}\right)^{-1}.$$

This is only true for one dimensional transformations.

5.2.5 Inverse Transformation

In general, for a transformation

$$(\xi_1, \xi_2, \ldots, \xi_n) \xrightarrow{T} (x_1, x_2, \ldots, x_n)$$

$$x_i = x_i(\xi_1, \xi_2, \ldots, \xi_n), \quad i = 1, 2, \ldots, n.$$

Then consider

$$J_{ij} = \frac{\partial x_i}{\partial \xi_j}, \quad \tilde{J}_{ij} = \frac{\partial \xi_i}{\partial x_j}.$$

For

$$\frac{\partial x_i}{\partial x_j} = \sum_{k=1}^{n} \frac{\partial x_i}{\partial \xi_k} \frac{\partial \xi_k}{\partial x_j} = \sum_{k=1}^{n} J_{ik} \tilde{J}_{kj} = \delta_{ij}.$$

This shows that the Jacobian matrix of the inverse transformation $\tilde{J}$ is indeed the matrix inverse of the Jacobian matrix J of the original transformation.

$$\implies J \cdot \tilde{J} = I.$$

5.3 Linear Algebra

5.3.1 Vector Space

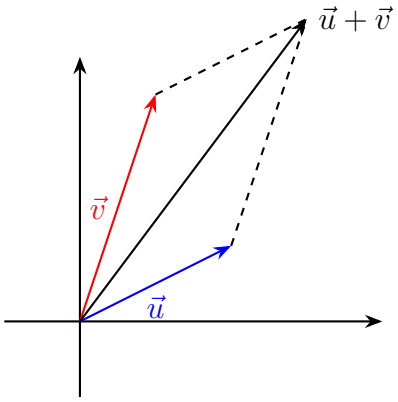
Definition 5.6

A vector space V over $\mathbb{R}$(or $\mathbb{C}$) is a set n.t.,

$$\forall \vec{u},\, \vec{v},\, \vec{w} \in V, \quad \forall \alpha, \beta \in \mathbb{R} \text{ or } \mathbb{C}$$

1. $\vec{u} + \vec{v} \in V$.
2. $\vec{u} + (-\vec{u}) = \vec{0} \in V$.
3. $\exists\, \vec{0} \in V$ such that $\vec{u} + \vec{0} = \vec{u}$.
4. $\exists\, (-\vec{u}) \in V$ such that $\vec{u} + (-\vec{u}) = \vec{0}$.
5. $\alpha(\beta\vec{u}) = (\alpha\beta)\vec{u}$
6. $\alpha(\vec{u} + \vec{v}) = \alpha\vec{u} + \alpha\vec{v}$.
7. $(\alpha + \beta)\vec{u} = \alpha\vec{u} + \beta\vec{u}$.

Addition



5.3.2 Linearly Independent

Definition 5.7

A set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} \subseteq V$ is said to be linearly independent if

$$\sum_{i=1}^{k} \alpha_i \vec{v}_i = \vec{0} \iff \alpha_i = 0 \quad i = 1, 2, \ldots, k.$$

Otherwise, the set is linearly dependent.

5.3.3 Span and Basis

Definition 5.8

The span of a set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} \subseteq V$ is defined as

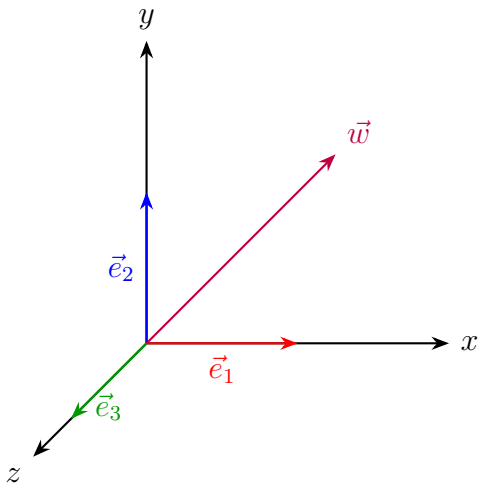
$$\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} = \left\{ \vec{v} = \sum_{i=1}^{n} \alpha_i \vec{v}_i \in V \mid \alpha_i \in \mathbb{R} \text{ or } \mathbb{C} \right\}$$

The span is a set of vector can be written as all possible linear combinations of the given vectors, and it not necessarily linearly independent and equals to the whole vector space only if the vectors form a basis.

Definition 5.9

A set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} \subseteq V$ is said to be a basis of V if it is linearly independent and

$$\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} = V.$$



So for $\{e_1, e_2, \ldots, e_n\}$ is the basis of V, every vector $\vec{v} \in V$ can be uniquely written as

$$\vec{v} = \sum_{i=1}^{n} v_i e_i,$$

where $v_i \in \mathbb{R}$ or $\mathbb{C}$ for $i = 1, 2, \ldots, n$.

Remark

The v_i are called the coordinates/ components/ entries of $\vec{v}$ with respect to the basis $\{e_1, e_2, \ldots, e_n\}$.

Same vector but different basis will have different coordinates/components/entries.

5.3.4 Inner Product

Definition 5.10

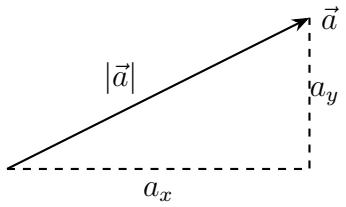
An inner product on a vector space V is a function that assigns a scalar to each pair of vectors in V

$$\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{R} \text{ or } \mathbb{C}$$
$$(\vec{u}, \vec{v}) \mapsto \langle \vec{u}, \vec{v} \rangle.$$

And satisfies the following properties

1. $\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$
2. $\langle \alpha\vec{u} + \beta\vec{w}, \vec{v} \rangle = \underbrace{\alpha\langle \vec{u}, \vec{v} \rangle + \beta\langle \vec{w}, \vec{v} \rangle}_{\in \mathbb{R} \text{ or } \mathbb{C}}$
3. $\langle \vec{v}, \vec{v} \rangle \geq 0$

$$\langle \vec{v}, \vec{v} \rangle = 0 \iff \vec{v} = \vec{0}$$



Theorem 5.1

Cauchy-Schwarz Inequality

For any vectors $\vec{u}, \vec{v} \in V$,

$$|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \cdot \|\vec{v}\|,$$

where $\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$ is the norm induced by the inner product.

Proof. Consider the vector

$$\vec{w} = \vec{u} + \lambda\vec{v}\, \lambda \in \mathbb{R}$$

then

$$\begin{aligned} \|\vec{w}\|^2 &= \langle \vec{w}, \vec{w} \rangle \\ &= \langle \vec{u} + \lambda\vec{v}, \vec{u} + \lambda\vec{v} \rangle \\ &= \langle \vec{u}, \vec{u} \rangle + \lambda\langle \vec{u}, \vec{v} \rangle + \lambda\langle \vec{v}, \vec{u} \rangle + \lambda^2\langle \vec{v}, \vec{v} \rangle \\ &= |\vec{u}|^2 + 2\lambda\langle \vec{u}, \vec{v} \rangle + \lambda^2 |\vec{v}|^2 \geq 0 \end{aligned}$$

Is a polynomial in λ and must be non-negative for all $\lambda \in \mathbb{R}$. Therefore, its discriminant must be less than or equal to zero

$$4|\langle \vec{u}, \vec{v} \rangle|^2 - 4\,|\vec{u}|^2\,|\vec{v}|^2 \leq 0$$

which implies

$$|\langle \vec{u}, \vec{v} \rangle|^2 \leq |\vec{u}|^2 \, |\vec{v}|^2$$

Taking the square root of both sides, we obtain the Cauchy-Schwarz inequality

$$|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \cdot \|\vec{v}\|.$$

□

5.3.5 Orthonormal Basis

Definition 5.11

A set of vectors $\{\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n\}$ in an inner product space V is called an *orthonormal basis* if

$$\langle \vec{e}_i, \vec{e}_j \rangle = \delta_{ij}, \quad \text{for all } i, j$$

1. $\langle \vec{e}_i, \vec{e}_j \rangle = 0$ for all $i \neq j$
2. $\langle \vec{e}_i, \vec{e}_i \rangle = 1$ for all i

For any vector $\vec{v} \in V$, it can be expressed as a linear combination of the orthonormal basis vectors

$$\vec{v} = \sum_{i=1}^{n} v_i \vec{e}_i.$$

Where the coefficients v_i are given by the inner products with the basis vectors

$$\begin{aligned} v_k &= \langle \vec{v}, \vec{e}_k \rangle \\ &= \left\langle \sum_{i=1}^{n} v_i \vec{e}_i, \vec{e}_k \right\rangle \\ &= \sum_{i=1}^{n} v_i \langle \vec{e}_i, \vec{e}_k \rangle \\ &= \sum_{i=1}^{n} v_i \delta_{ik} = v_k. \end{aligned}$$

We can use the orthonormality of the basis vectors to compute the norm of $\vec{v}$ (Parsing Theorem)

$$\begin{aligned} |\vec{v}|^2 &= \langle \sum_{i=1}^{n} v_i \vec{e}_i, \sum_{j=1}^{n} v_j \vec{e}_j \rangle \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} v_i \overline{v_j} \langle \vec{e}_i, \vec{e}_j \rangle \\ &= \sum_{i=1}^{n} v_i^2 \ \in \mathbb{R}_\lambda. \end{aligned}$$

6 Lecture 6: Dynamic in One Dimension

6.1 Particles

Definition 6.1

A *particle* is a particle object with no internal structure, characterized only by its mass and position in space.

Definition 6.2

The *state* of a system of particles is described by the positions and velocities of all the particles.

$$(x(t),\ v(t))\,, \quad x(t),\ v(t) \in \mathbb{R}.$$

$$v(t) = \frac{dx(t)}{dt}.$$

All observable state are function of the *state vector*

$$\vec{S}(t) = (x(t),\ v(t)) \ \in \mathbb{R}^2.$$

Remark

The state vector is made of observable quantities, namely the position and velocity of the particle.

Definition 6.3

Acceleration of a particle is the time derivative of its velocity.

$$a(t) = \frac{dv(t)}{dt} = \frac{d^2x(t)}{dt^2}.$$

6.2 Newtonian Dynamics

Is the system involve time, is the Newton's Laws.

Theorem 6.1

Inertia

There *exist* inertial frames in which a particle not subject to external force remains at rest or moves with constant velocity.

Theorem 6.2

Equation of Motion

In an inertial frame

$$\vec{F} = \frac{d\vec{p}}{dt},\ m = \text{const} \implies \vec{F} = m\vec{a}.$$

F is the totle froce acting on the particle.

$$\vec{p} = m\vec{v}.$$

And for unit, in the SI system,

$$[P] = MLT^{-1}, \quad [F] = MLT^{-2} = N.$$

Theorem 6.3

Action and Reaction

For every action, there is an equal and opposite reaction.

$$\vec{F}_{12} + \vec{F}_{21} = 0.$$

Equation of motion is a second-order ordinary differential equation for the position of the particle,

$$F = ma = m\ddot{x}.$$

To solve it, we need two initial conditions, typically the initial position and initial velocity of the particle.

$$x(0) = x_0, \quad v(0) = v_0.$$

If the notation is known for all t, it implies the deterministic time evolution of the system.

Example

for a particle of m under gravtional force,

$$F = -mg,$$

the equation of motion is

$$m\ddot{x} = -mg \implies \ddot{x} = -g.$$

With initial conditions

$$x(0) = x_0, \quad v(0) = v_0,$$

the solution is

$$x(t) = x_0 + v_0 t - \frac{1}{2}gt^2.$$

for the unit, in the SI system,

$$[g] = LT^{-2}, \quad [gt^2] = LT^{-2}T^2 = L.$$

Example

Linear restoring force, a spring with Hooke's law,

$$F = -kx,$$

the equation of motion is

$$m\ddot{x} = -kx,$$
$$MLT^{-2} = [k]L \implies [k] = MT^{-2}.$$

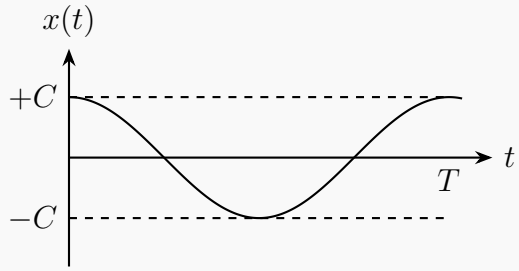
Rewrite the equation of motion as

$$\ddot{x} = -\frac{k}{m}x.$$

Because the dimension of $\frac{k}{m}$ is T^{-2}, it has the dimension of the square of angular frequency.

$$\ddot{x} = -\frac{k}{m}x \implies \ddot{x} + \omega^2 x = 0,$$
$$[\omega] = T^{-1}.$$

- m is the property of the system.
- ω is the "scale" of the system's oscillation.



Example

particle on the slope, with angle θ to the horizontal and

- $\vec{N}$ is the normal force
- $-m\vec{g}$ is the gravitational force

Reduced to one dimension,

$$\begin{cases} \perp & m\ddot{x} = -mg\sin\theta, \\ \parallel & \ddot{x} = -g\sin\theta. \end{cases}$$

Check limiting cases

- $\theta = 0$: $\ddot{x} = 0$, no acceleration along the slope.
- $\theta = \frac{\pi}{2}$: $\ddot{x} = -g$, free fall.

6.3 Energy

Definition 6.4

Work done by a force $F(\vec{x})$ along a displacement $\vec{dr}$ is defined as

$$W_{A\to B} = \int_{x_A}^{x_B} \vec{F} \cdot d\vec{r},$$

$$[W] = ML^2T^{-2}$$

Definition 6.5

Kinetic Energy of a particle with mass m and velocity $\vec{v}$ is defined as

$$T = \frac{1}{2}mv^2,$$

$$[T] = ML^2T^{-2}.$$

Remark

Energy has the dimension of ML^2T^{-2}, same as work.

$$[T] = ML^2T^{-2} = [W]$$

So work is a form of energy.

Proof. Consider a particle moving along a path from point A to point B under the influence of a force $F(x)$.

The work done by the force is

$$W_{A\to B} = \int_{t_A}^{t_B} dt\, \dot{x}(t)F(x(t)) = \int_{t_A}^{t_B} dt\, \frac{P}{m}(\dot{p}) = \int_{t_A}^{t_B} dt\, \frac{d}{dt}\left(\frac{P^2}{2m}\right) = T_B - T_A.$$

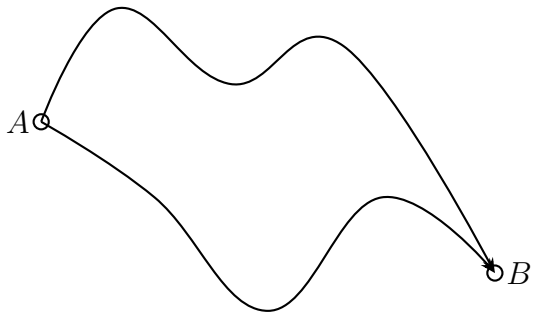
□

Definition 6.6

A force $\vec{F}$ is called conservative if it can be written as the negative gradient of a potential energy function $V(\vec{r})$:

$$\vec{F} = -\nabla V(\vec{r}).$$

$W_{A\to B}$ does not depend on the path taken, only on the endpoints A and B.



Any force on 1-dimensional motion can be written as the gradient of a potential energy function,

$$F(x) = -\frac{dV(x)}{dx}.$$

And the workdone by a conservative force is equal to the negative change in potential energy,

$$W_{A\to B} = V(\vec{r}_A) - V(\vec{r}_B).$$

6.4 Conservation Laws

Theorem 6.4

Conservation of Energy

$$E = \frac{p^2}{2m} + V(x) \text{ is conserved.}$$

Proof.

$$\frac{dE}{dt} = \frac{d}{dt}\left(\frac{p^2}{2m} + V(x)\right) = \frac{p}{m}\left(\dot{p} + \frac{dV}{dx}\right) = 0.$$

□

For 1-dimensional motion under a conservative force,

$$\frac{1}{2}\left(\frac{dx}{dt}\right)^2 + V(x) = E \implies \left(\frac{dx}{dt}\right) = \pm\sqrt{2(E - V(x))}.$$

And the time evolution can be obtained by integrating the equation of motion,

$$dt = \frac{dx}{\pm\sqrt{2(E - V(x))}} \implies t = \pm\int_{x_0}^{x} \frac{d\xi}{\pm\sqrt{2(E - V(\xi))}}.$$

Boundary condition as below,

$$x(t_0) = x_0.$$

We need to find the $x(t)$,

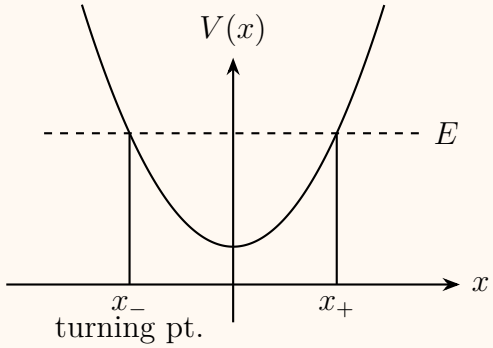
$$\int_0^t dt' = \int_{x_0}^{x(t)} \frac{d\xi}{\pm\sqrt{2(E - V(\xi))}}$$

This integral equation defines $x(t)$ implicitly as a function of time.

Remark

The energy is always larger than or equal to the potential energy,

$$E = \frac{p^2}{2m} + V(x) \geq V(x).$$



At turning points, the kinetic energy is zero and the particle changes direction. And at the minimum of the potential energy, the particle have highest kinetic energy.

Theorem 6.5

Conservation of Momentum

$$p = \sum_i p_i = \sum_i m_i \frac{dx_i}{dt}.$$

is conserved if the total external force is zero.

Proof.

$$\dot{p} = \sum \dot{p}_i = \sum \left(F_i^{\text{ext}} + \sum_{g \neq i} F_{ig}^{\text{int}} \right) = \sum_i F_i^{\text{ext}} + \sum_{i,j}{}' F_{ij}^{\text{int}}.$$

The initial force cancelled by the internal forces due to Newton's third law, leaving only the external forces to change the total momentum.

$$\dot{p} = \sum_i F_i^{\text{ext}} = F^{\text{ext}}.$$

So if

$$F^{\text{ext}} = 0,$$

then

$$\dot{p} = 0 \implies p = \text{constant}.$$

□

Internal forces can be described by a potential function $V(x_1, x_2, \ldots, x_n)$,

$$F_i^{\text{int}} = -\frac{\partial V}{\partial x_i}.$$

Internal potential energy are invariant under the translation of the entire system,

$$V(x_1 + \epsilon, x_2 + \epsilon, \ldots, x_n + \epsilon) = V(x_1, x_2, \ldots, x_n),$$

for any constant vector ϵ.

$$\frac{d}{d\epsilon}V(x_1+\epsilon, x_2+\epsilon, \ldots, x_n+\epsilon)|_{\epsilon=0} = \sum_i \frac{d}{d\epsilon}\frac{\partial V}{\partial x_i}.$$

$$\sum_i \frac{\partial V}{\partial x_i} = 0.$$

Remark

Conservation of momentum follows from the invariance of the internal potential energy under space translation.

It is the part of Noether's theorem that associates symmetries of the system with conservation laws.

6.5 Oscillators

6.5.1 Harmonic Oscillator

Back to Hooke's law, the force on a mass attached to a spring is proportional to its displacement,

$$F = -kx,$$

where k is the spring constant.

$$m\ddot{x} + kx = 0 \implies \ddot{x} + \omega^2 x = 0,$$

The general solution to this differential equation is

$$\begin{aligned} x(t) &= \alpha e^{i\omega t} + \beta e^{-i\omega t} \\ &= (\alpha+\beta)\cos(\omega t) + i(\alpha-\beta)\sin(\omega t). \end{aligned}$$

where α and β are constants determined by the initial conditions.

$$x(t) = A\cos(\omega t) + B\sin(\omega t).$$

We can solve for the constants A and B,

$$\begin{cases} \alpha+\beta = A, \\ i(\alpha-\beta) = B. \end{cases} \implies \begin{cases} \alpha = \frac{A-iB}{2}, \\ \beta = \frac{A+iB}{2}. \end{cases}$$

So for a real solution, the constants α and β must be complex conjugates of each other,

$$\alpha = \beta^*$$

And $x(t)$ is periodic with period

$$T = \frac{2\pi}{\omega}.$$

For kinetic energy,

$$K = \frac{1}{2}m\dot{x(t)}^2 = \frac{1}{2}m\omega^2 C^2 \sin^2(\omega t - \phi),$$

For potential energy,

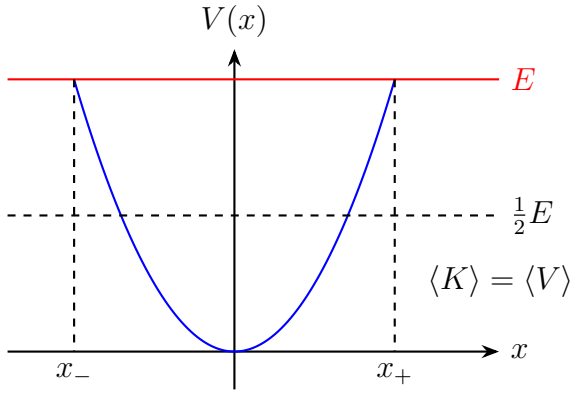
$$U = \frac{1}{2}kx(t)^2 = \frac{1}{2}kC^2\cos^2(\omega t - \phi),$$

The total energy is the sum of kinetic and potential energies,

$$\begin{aligned} E &= K + U \\ &= \frac{1}{2}m\omega^2 C^2 \sin^2(\omega t - \phi) + \frac{1}{2}kC^2\cos^2(\omega t - \phi) \\ &= \frac{1}{2}m\omega^2 C^2. \end{aligned}$$

This shows that the total energy of a harmonic oscillator remains constant over time, as expected for a conservative system.

$$\langle K \rangle = \langle V \rangle = \frac{1}{2}E$$



Example

Exercise, Intergrate

$$\frac{dx}{dt} = \pm\sqrt{\frac{2}{m}\left(E - \frac{1}{2}m\omega^2 x^2\right)}$$

For boundary conditions, we have

$$x(0) = x_0$$

EOM

$$\text{linear in } x \iff \text{Superposition principle holds.}$$

Remark

Every system that close to a stable equilibrium can be approximated as a harmonic oscillator near that equilibrium.

$$V(x) = V(x_0) + (x - x_0)V'(x_0) + \frac{1}{2}(x - x_0)^2 V''(x_0) + \dots$$

6.5.2 Damped Harmonic Oscillator

Add a time-dependent damping term to the harmonic oscillator equation, we get the damped harmonic oscillator,

$$m\ddot{x} + kx + b\dot{x} = 0.$$

There is a friction opposite to the direction of motion, represented by the damping term $b\dot{x}$.

$$\ddot{x} + 2\gamma\dot{x} + \omega^2 x = 0,$$

$$\gamma = \frac{b}{2m}, \quad [\gamma] = T^{-1}$$

Then we assume a solution of the form

$$x(t) = e^{\lambda t},$$

Substituting $x(t) = e^{\lambda t}$ into the differential equation, we get the characteristic equation,

$$\lambda^2 + 2\gamma\lambda + \omega^2 = 0.$$

Solve the characteristic equation,

$$\lambda = -\gamma \pm \sqrt{\gamma^2 - \omega^2}.$$

Depending on the discriminant $\gamma^2 - \omega^2$, we have three cases.

- Underdamped Case

  $$\gamma < \omega, \quad \gamma/\omega \leq 1$$

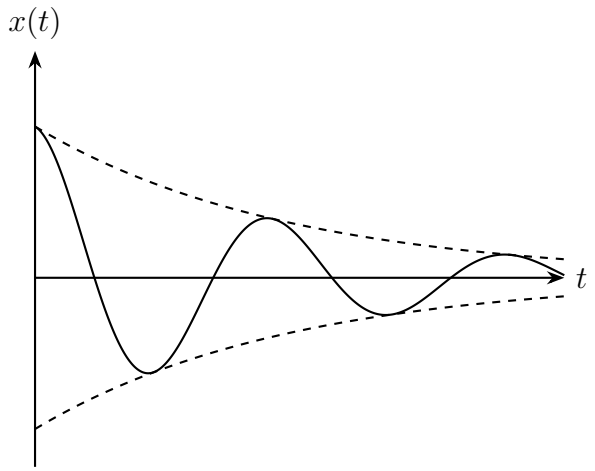
  So the characteristic roots are complex conjugates,

  $$\lambda = -\gamma \pm \sqrt{\gamma^2 - \omega^2} = -\gamma \pm i\sqrt{\omega^2 - \gamma^2}$$

  The general solution for the underdamped case is

  $$x(t) = e^{-\gamma t}\left(A\cos(\sqrt{\omega^2 - \gamma^2}\, t) + B\sin(\sqrt{\omega^2 - \gamma^2}\, t)\right),$$

  $$x(t) = e^{-\gamma t} C\cos(\sqrt{\omega^2 - \gamma^2}\, t + \phi).$$



- Critically Damped Case

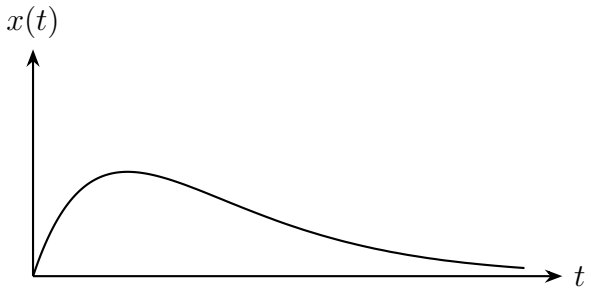
  $$\gamma = \omega$$

  The characteristic roots are real and equal,

  $$\lambda = -\gamma$$

  The general solution for the critically damped case is

  $$x(t) = (C_1 + C_2 t)e^{-\gamma t}$$



- Overdamped Case

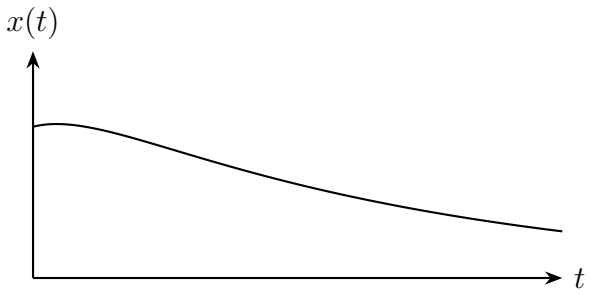
$$\gamma > \omega$$

The characteristic roots are real and distinct,

$$\lambda = -\gamma \pm \sqrt{\gamma^2 - \omega^2}$$

The general solution for the overdamped case is

$$x(t) = C_1 e^{\left(-\gamma + \sqrt{\gamma^2-\omega^2}\right)t} + C_2 e^{\left(-\gamma - \sqrt{\gamma^2-\omega^2}\right)t}$$



7 Lecture 7: Hamiltonian and Oscillations

7.1 Forced Damped Oscillations

Definition 7.1

$$m\ddot{x} = -b\dot{x} - kx + F_0 \cos(\omega t)$$

rewriting the equation in standard form,

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos(\omega t) = Re\left(\frac{F_0}{m}e^{i\omega t}\right)$$

In order to find the most general solution, ignoring the right hand side, and fix the solution with the particular solution later.

$\gamma < \omega_0$ **(Underdamped Case)**

We have the homogeneous solution as

$$x_h(t) = \alpha e^{-\gamma t} e^{i\omega_\alpha t} + \beta e^{-\gamma t} e^{-i\omega_\alpha t},$$

and

$$\omega_\alpha = \sqrt{\omega_0^2 - \gamma^2}.$$

Inhomogeneous solution can be found by assuming a particular solution of the form

$$x_p(t) = Ae^{i\omega t}.$$

Substituting into the differential equation gives

$$(-\omega^2 + 2i\gamma\omega + \omega_0^2)Ae^{i\omega t} = \frac{F_0}{m}e^{i\omega t}.$$

Hence,

$$A = \frac{F_0/m}{\omega_0^2 - \omega^2 + 2i\gamma\omega}.$$

Change the denominator into polar form,

$$\omega_0^2 - \omega^2 + 2i\gamma\omega = |\psi|e^{i\phi} = \sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}e^{i\phi},$$

$$\phi = \arctan\left(\frac{2\gamma\omega}{\omega_0^2 - \omega^2}\right).$$

Therefore, A can be written as

$$A = \frac{F_0/m}{|\psi|e^{i\phi}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}e^{-i\phi}.$$

the particular solution is

$$x_p(t) = Ae^{i\omega t} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}e^{i(\omega t - \phi)}.$$

$$Re\left[\frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}e^{i(\omega t - \phi)}\right] = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}\cos(\omega t - \phi).$$

Complete solution is the sum of homogeneous and particular solutions,

$$x(t) = x_h(t) + x_p(t).$$

We have the period of the driving force as

$$T = \frac{2\pi}{\omega}, \quad \omega T = 2\pi.$$

When we average energy of the system over one period, we have

$$\Delta E = \int dx F_b = \int dt \frac{dx}{dt}(-b\dot{x}) = -b\int_0^T \dot{x}^2 dt.$$

From the particular solution $x_p(t) = |A|e^{i(\omega t + \phi)}$, we have

$$\dot{x}(t) = -\omega|A|\sin(\omega t + \phi).$$

Therefore, set $\tau = \omega t$,

$$\begin{aligned} W_b &= -b \int dt \omega^2 |A|^2 \sin^2(\omega t + \phi) \\ &= -b \int_0^{2\pi} \frac{d\tau}{\omega} \omega^2 |A|^2 \sin^2(\tau + \phi) \\ &= -b\omega |A|^2 \int_0^{2\pi} \sin^2(\tau + \phi) d\tau \\ &= -b\omega |A|^2 \pi. \end{aligned}$$

Remark

The driving force is not conservative, and the work done by the damping force over one period is negative, indicating energy dissipation.

$$\begin{aligned} W_d &= \int dx\, F_d \\ &= \int dt\, \frac{dx}{dt} F_d \\ &= \int_0^T dt\, \dot{x} F_0 \cos(\omega t) \\ &= -\int_0^T dt\, \omega |A| \sin(\omega t + \phi) F_0 \cos(\omega t) \\ &= -\omega |A| F_0 \int_0^T \sin(\omega t + \phi) \cos(\omega t)\, dt \\ &= W_b. \end{aligned}$$

7.2 Hamiltonian Mechanics

7.2.1 Hamiltonian Formula

State of a system

$$(x, \dot{x}) \to (q, p)$$

where q is the generalized coordinate and p is the conjugate momentum,

$$\begin{cases} q = x, \\ p = m\dot{x}. \end{cases}$$

We have the hamiltonian

$$H = T + V = \frac{p^2}{2m} + V(q),$$

where $V(q)$ is the potential energy as a function of the generalized coordinate q.

Theorem 7.1

Hamilton's equations describe the time evolution of the generalized coordinates and conjugate momenta in a Hamiltonian system. We can write it as

$$\begin{cases} \dot{q} = \frac{\partial H}{\partial p}, \\ \dot{p} = -\frac{\partial H}{\partial q}. \end{cases}$$

with

$$H = T + V.$$

We can compute the partial derivatives of the hamiltonian to obtain the equations of motion,

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m},$$
$$\ddot{q} = \frac{d}{dt}\frac{p}{m} = \frac{\dot{p}}{m} = -\frac{1}{m}\frac{\partial V}{\partial q}.$$

So we get the newtonian form of the equations of motion,

$$m\ddot{q} = -\frac{\partial V}{\partial q}.$$

Conservation of energy take place,

$$\frac{d}{dt}H(q(t), p(t)) = \frac{\partial H}{\partial q}\dot{q} + \frac{\partial H}{\partial p}\dot{p} = \frac{\partial H}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial H}{\partial p}\frac{\partial H}{\partial q} = 0.$$

Example

A free particle $V(q) = 0$. The Hamiltonian is

$$H = \frac{p^2}{2m}.$$

Applying Hamilton's equations, we get

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial H}{\partial q} = 0.$$

Therefore, the particle moves with constant velocity, as expected for a free particle.

7.2.2 Harmonic Oscillator

Harmonic oscillator potential is given by

$$V(q) = \frac{1}{2}m\omega^2 q^2.$$

Computing the derivative of the potential, we get

$$V'(q) = \frac{\partial V}{\partial q} = m\omega^2 q.$$

When we look at energy conservation, we have

$$E = T + V = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2.$$

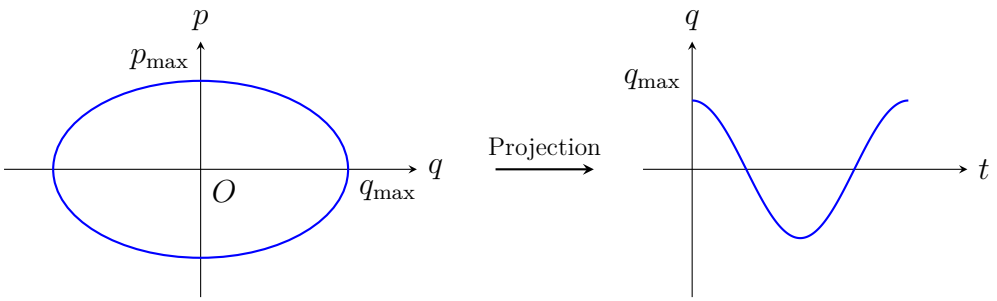
The harmonic oscillator in the p-q plane is a ellipse, representing the constant energy contours of the system.

$$q_{\max} = \sqrt{\frac{2E}{m\omega^2}},$$
$$p_{\max} = \sqrt{2mE}.$$

In x-t space, the motion of the harmonic oscillator is described by

$$q(0) = q_{\max},$$
$$\dot{q}(0) = 0.$$

so there is a transformation from the p-q plane to the x-t plane.



The area enclosed by the ellipse in the p-q plane is proportional to the energy of the system.

$$\begin{aligned}\text{Area} &= \pi q_{\max} p_{\max} \\ &= \pi\sqrt{\frac{2E}{m\omega^2}}\sqrt{2mE} \\ &= \frac{2\pi E}{\omega} \\ &= E \cdot T.\end{aligned}$$

7.2.3 Perturbation Theory

If we consider a small perturbation in the energy,

$$E_0 = H(q_0, p_0)$$

Consider the same system with a new energy

$$E = E_0 + \epsilon E_1 + \dots.$$

Small so that the perturbation is weak, we have,

$$\epsilon \ll 1.$$

Evolution on a "new" orbit $(q(t), p(t))$,

$$q(t) = q_0 + \epsilon q_1(t) + \dots,$$
$$p(t) = p_0 + \epsilon p_1(t) + \dots$$

The dots mean higher-order terms in the perturbation expansion.

$$\cdots \sim \mathcal{O}(\epsilon^2).$$

Apply the Hamilton's equations to the perturbed orbit,

$$\begin{aligned}\dot{q}(t) &= \dot{q}_0(t) + \epsilon \dot{q}_1(t), \\ &= \frac{\partial H}{\partial p}(q_0, p_0) + \left.\frac{\partial^2 H}{\partial q \partial p}\right|_0 \epsilon q_1 + \left.\frac{\partial^2 H}{\partial p^2}\right|_0 \epsilon p_1\end{aligned}$$

So we have, for $\mathcal{O}(\epsilon)$,

$$\begin{aligned}\dot{q}_1(t) &= \left.\frac{\partial^2 H}{\partial q \partial p}\right|_0 q_1 + \left.\frac{\partial^2 H}{\partial p^2}\right|_0 p_1, \\ \dot{p}_1(t) &= -\left.\frac{\partial^2 H}{\partial q^2}\right|_0 q_1 - \left.\frac{\partial^2 H}{\partial q \partial p}\right|_0 p_1.\end{aligned}$$

Energy,

$$E = H(q(t), p(t)) \implies E_0 + \epsilon E_1 = H(q_0 + \epsilon q_1, p_0 + \epsilon p_1)$$

Expanding the Hamiltonian to first order in ϵ,

$$H(q_0 + \epsilon q_1, p_0 + \epsilon p_1) = H(q_0, p_0) + \epsilon \left(\left.\frac{\partial H}{\partial q}\right|_0 q_1 + \left.\frac{\partial H}{\partial p}\right|_0 p_1 \right) + \mathcal{O}(\epsilon^2).$$

Comparing with $E_0 + \epsilon E_1$, we get

$$\begin{aligned}E_1 &= \left.\frac{\partial H}{\partial q}\right|_0 q_1 + \left.\frac{\partial H}{\partial p}\right|_0 p_1 \\ &= \dot{q}_0 p_1 - \dot{p}_0 q_1\end{aligned}$$

Remark

$$H(q, p) = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 q^2$$

The initial conditions are

$$\begin{aligned}q(0) &= q_0, \\ p(0) &= 0\end{aligned}$$

We can solve the equations of motion to get the time evolution of $q(t)$ and $p(t)$.

$$q(t) = q_0 \cos(\omega t)$$

$$H(p_0, q_0) = E_0 = \frac{1}{2} m\omega^2 q_0^2$$

$$p(t) = -m\omega q_0 \sin(\omega t)$$

This is an elliptical motion in phase space, with the semi-major axis q_0 and semi-minor axis $m\omega q_0$.

Consider the energy of simple harmonic oscillator,

$$E = E_0 + \epsilon E_1.$$

And the time evolution of $q(t)$ and $p(t)$ can be expressed as (where $q_1(t)$ and $p_1(t)$ need to be determined),

$$q(t) = q_0(t) + \epsilon q_1(t),$$
$$p(t) = p_0(t) + \epsilon p_1(t).$$

We have the matrix form of the linearized equations of motion,

$$\frac{d}{dt}\begin{pmatrix} q_1 \\ p_1 \end{pmatrix} = \begin{pmatrix} \frac{\partial^2 H}{\partial q \partial p} & \frac{\partial^2 H}{\partial p^2} \\ -\frac{\partial^2 H}{\partial q^2} & -\frac{\partial^2 H}{\partial p \partial q} \end{pmatrix}_{(q_0, p_0)} \begin{pmatrix} q_1 \\ p_1 \end{pmatrix}.$$

For this case of the harmonic oscillator, where

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2,$$

the matrix of the linearized equations of motion becomes,

$$\begin{pmatrix} \dot{q}_1 \\ \dot{p}_1 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{m} \\ -m\omega^2 & 0 \end{pmatrix} \begin{pmatrix} q_1 \\ p_1 \end{pmatrix}.$$

So we have the linear differential equations by matrix multiplication,

$$\dot{q}_1 = \frac{1}{m}p_1,$$
$$\dot{p}_1 = -m\omega^2 q_1,$$
$$m\ddot{q}_1 = -m\omega^2 q_1.$$

We get the equation of a simple harmonic oscillator with the same frequency ω,

$$\ddot{q}_1 + \omega^2 q_1 = 0.$$

Then the solution for $q_1(t)$ and $p_1(t)$ can be written as

$$\begin{cases} q_1(t) = A\cos(\omega t) + B\sin(\omega t), \\ p_1(t) = -m\omega A\sin(\omega t) + m\omega B\cos(\omega t) = m\omega\left[-A\sin(\omega t) + B\cos(\omega t)\right]. \end{cases}$$

Substitute it into the equation of E_1,

$$E_1 = -q_1\dot{p}_0 + p_1\dot{q}_0.$$

We have

$$\begin{aligned} E_1 &= \sqrt{2m\omega^2 E_0}\left[C_1\cos^2(\omega t) + C_2\cos(\omega t)\sin(\omega t) + C_1\sin^2(\omega t) - C_2\sin(\omega t)\cos(\omega t)\right] \\ &= \sqrt{2m\omega^2 E_0}\left[C_1(\cos^2(\omega t) + \sin^2(\omega t))\right] \\ &= C_1\sqrt{2m\omega^2 E_0}. \end{aligned}$$

We use the initial condition to determine C_2,

$$p_1(0) = 0 \implies C_2 = 0.$$

In the newtonian picture, the position of the particle is given by

$$q(t) = q_0(t) + \epsilon q_1(t) = \underbrace{\sqrt{\frac{2E_0}{m\omega^2}}}_{q_0(t)} \left(1 + \epsilon \frac{E_1}{2E_0}\right) \cos(\omega t).$$

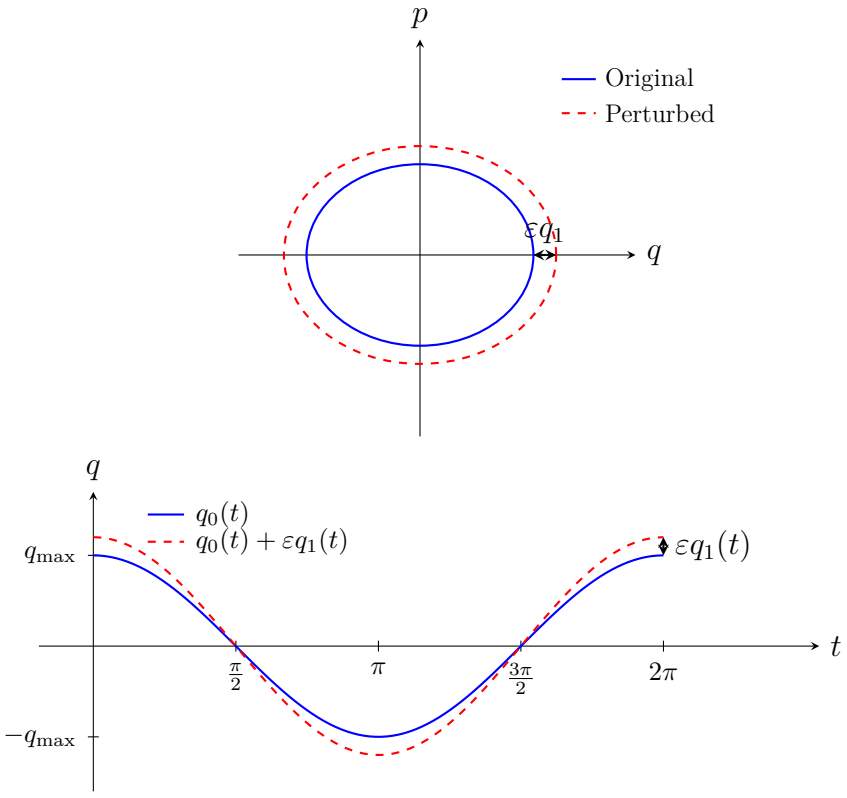
is a cosine function with a modified amplitude due to the perturbation,

$$q(t) = \sqrt{\frac{2E}{m\omega^2}} \cos(\omega t) = \sqrt{\frac{2(E_0)}{m\omega^2}} \sqrt{1 + \frac{\epsilon E_1}{E_0}} \cos(\omega t).$$

In this equation, we use the expansion

$$\sqrt{1 + \frac{\epsilon E_1}{E_0}} = 1 + \frac{1}{2}\frac{\epsilon E_1}{E_0} + \mathcal{O}(\epsilon^2).$$

The following phase space and time evolution diagrams, shows that the perturbation modifies the amplitude of the oscillation. Both in phase space and in the newtonian picture.



8 Lecture 8: Dynamics In Two/Three Dimensions

8.1 Position

Definition 8.1

The position of a particle in two or three dimensions is described by a vector

$$\mathbf{r}(t) = \vec{r}(t).$$

Which specifies the coordinates of the particle as functions of time.

$$\begin{aligned}\vec{r}(t) &= x(t)\vec{e_x} + y(t)\vec{e_y} + z(t)\vec{e_z} \\ &= x_1(t)\vec{e_1} + x_2(t)\vec{e_2} + x_3(t)\vec{e_3} \\ &= \sum_{i=1}^{3} x_i(t)\vec{e_i}\end{aligned}$$

The $\vec{e_i}$ is orthonormal basis vector in the corresponding coordinate system.

$$\vec{e_i} \cdot \vec{e_j} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

And $x_i(t)$ are the components of the position vector along the corresponding basis vectors, which are called the Cartesian coordinates in the respective coordinate system.

$$\vec{r}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix}$$

This associates the choice of basis vectors with the components of the position vector,

$$\begin{aligned}\vec{r}(t) &= \sum_{i=1}^{3} x_i(t)\vec{e_i} \\ &= \sum_{i=1}^{3} x_i'(t)\vec{e_i'}\end{aligned}$$

8.2 Velocity

Definition 8.2

The velocity of a particle in two or three dimensions is the time derivative of its position vector:

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt}.$$

Consider a small time interval Δt,

$$\Delta\vec{r}(t) = \vec{r}(t + \Delta t) - \vec{r}(t).$$

The velocity can then be expressed as the limit of the average velocity,

$$\vec{v}(t) = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}(t)}{\Delta t}.$$

And then when we consider the basis vectors, the velocity can be expressed in components as

$$\begin{aligned}\vec{v}(t) &= \frac{d}{dt}\left(\sum_{i=1}^{3} x_i(t)\vec{e_i}\right) \\ &= \sum_{i=1}^{3} \frac{dx_i(t)}{dt}\vec{e_i} \\ &= \sum_{i=1}^{3} \dot{x_i}(t)\vec{e_i}\end{aligned}$$

8.3 Acceleration

Definition 8.3

The acceleration of a particle in two or three dimensions is the time derivative of its velocity vector:

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt}.$$

Consider a small time interval Δt,

$$\Delta \vec{v}(t) = \vec{v}(t + \Delta t) - \vec{v}(t).$$

The acceleration can then be expressed as the limit of the average acceleration,

$$\vec{a}(t) = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}(t)}{\Delta t}.$$

And then when we consider the basis vectors, the acceleration can be expressed in components as

$$\begin{aligned}\vec{a}(t) &= \frac{d}{dt}\left(\sum_{i=1}^{3} \dot{x_i}(t)\vec{e_i}\right) \\ &= \sum_{i=1}^{3} \frac{d^2x_i(t)}{dt^2}\vec{e_i} \\ &= \sum_{i=1}^{3} \ddot{x_i}(t)\vec{e_i}\end{aligned}$$

8.4 Polar Coordinate In The Plane

8.4.1 Polar Coordinates

In polar coordinates, the position vector of a particle in the plane can be expressed as

$$\vec{r}(t) = r(t)\vec{e_r}.$$

The unit vectors $\vec{e_r}$ and $\vec{e_\theta}$ satisfy the following properties,

$$\begin{cases} \vec{e_r} \cdot \vec{e_\theta} = 0, \\ |\vec{e_r}| = |\vec{e_\theta}| = 1. \end{cases}$$

Express in polar coordinates, the basis vectors can be written as

$$\begin{cases} \vec{e_r} = (\cos\theta, \sin\theta) = (\frac{x_1}{|\vec{r}|}, \frac{y}{|\vec{r}|}), \\ \vec{e_\theta} = (-\sin\theta, \cos\theta) = (-\frac{y}{|\vec{r}|}, \frac{x_1}{|\vec{r}|}) \end{cases}$$

And we can get the time derivatives of the basis vectors as

$$\begin{cases} \frac{d\vec{e_r}}{dt} = \dot{\theta}(-\sin\theta, \cos\theta) = \dot{\theta}\vec{e_\theta}, \\ \frac{d\vec{e_\theta}}{dt} = -\dot{\theta}(\cos\theta, \sin\theta) = -\dot{\theta}\vec{e_r}. \end{cases}$$

So the velocity of a particle in polar coordinates can be expressed as

$$\vec{v}(t) = \frac{d}{dt}(r(t)\vec{e_r}(t)) = \dot{r}\vec{e_r} + r\dot{\theta}\vec{e_\theta}(t).$$

Where the two terms represent the radial and tangential components of the velocity, respectively.

$$\begin{cases} \dot{r} & \text{is radial component of the velocity} \\ r\dot{\theta} & \text{is tangential component of the velocity.} \end{cases}$$

Then we can express the acceleration in polar coordinates as

$$\vec{a} = \frac{d\vec{v}}{dt} = (\ddot{r} - r\dot{\theta}^2)\vec{e_r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{e_\theta}.$$

The dimension of acceleration is

$$[a] = LT^{-2}$$

Same as velocity, the acceleration also has radial and tangential components, given by

$$\begin{cases} \ddot{r} - r\dot{\theta}^2 & \text{is radial component of the acceleration} \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} & \text{is tangential component of the acceleration.} \end{cases}$$

8.4.2 Circular Motion

The condition for circular motion is that the radial distance remains constant, i.e.,

$$\begin{aligned} r &= \text{ const}, \\ \dot{r} &= \ddot{r} = 0. \end{aligned}$$

The velocity and acceleration for circular motion can be simplified as

$$\begin{aligned} \vec{v} &= r\dot{\theta}\vec{e_\theta}, \\ \vec{a} &= -r\dot{\theta}^2\vec{e_r} + r\ddot{\theta}\vec{e_\theta}. \end{aligned}$$

the magnitude of the acceleration is

$$|\vec{a}| = \sqrt{(r\dot{\theta}^2)^2 + (r\ddot{\theta})^2}.$$

On the geometric side, we have

$$\begin{aligned} \vec{v} &= \vec{\omega} \times \vec{r}, \\ |\vec{v}| &= \omega r \sin\phi, \\ |\vec{\omega}| &= \omega = \dot{\theta}. \end{aligned}$$

8.5 Relative Motion

8.5.1 Uniform Translation

Consider two reference frames O and O', with position vectors $\vec{r}$ and $\vec{r}'$ respectively. O' moves with respect to O with a velocity $\vec{V}$ wrt O.

$$\begin{cases} \vec{r} = \vec{r}' + \vec{V}t, \\ \vec{v} = \vec{v}' + \vec{V}, \\ \vec{a} = \vec{a}'. \end{cases}$$

The Relation is Galilean transformation from the frame O' to the frame O. The result of transformatuon remains same dynamic because it remians the same motion equations in both frames.

$$\ddot{x} = \ddot{x}'.$$

8.5.2 Rotation

First, see the basis vectors of the rotating frame O' and how they change with time,

$$\frac{d\vec{e_i'}}{dt}\Big|_O = \vec{\omega} \times \vec{e_i'}.$$

We consider a geometric vector

$$\vec{A} = \sum_{i=1}^{D} A_i' \vec{e_i'}.$$

The time derivative of $\vec{A}$ in the frame O is

$$\begin{aligned} \frac{d\vec{A}}{dt}\Big|_O &= \sum_i \left[\frac{dA_i'}{dt}\Big|_O \vec{e_i'} + A_i' \frac{d\vec{e_i'}}{dt}\Big|_O \right] \\ &= \sum_i \left[\frac{dA_i'}{dt}\Big|_O \vec{e_i'} + A_i'(\vec{\omega} \times \vec{e_i'}) \right] \\ &= \sum_i \left[\dot{A}_i' \vec{e_i'} + A_i'(\vec{\omega} \times \vec{e_i'}) \right]. \end{aligned}$$

Consider O' rotating with angular velocity $\vec{\omega}$ relative to O

$$\frac{d\vec{A}}{dt}\Big|_{O'} = \sum_i \dot{A}_i' \vec{e_i'}.$$

The time derivative of $\vec{A}$ in the frame O' is

$$\begin{aligned}
\frac{d\vec{A}}{dt}\Big|_{O'} &= \sum_i \left[\frac{dA'_i}{dt}\Big|_{O'} \vec{e'_i} + A'_i \frac{d\vec{e'_i}}{dt}\Big|_{O'} \right] \\
&= \sum_i \left[\frac{dA'_i}{dt}\Big|_{O'} \vec{e'_i} + A'_i(\vec{\omega} \times \vec{e'_i}) \right] \\
&= \sum_i \left[\dot{A}'_i \vec{e'_i} + A'_i(\vec{\omega} \times \vec{e'_i}) \right] \\
&= \vec{\omega} \times \left(\sum_i A'_i \vec{e'_i} \right) \\
&= \vec{\omega} \times \vec{A}.
\end{aligned}$$

Consider the velocity vector $\vec{v}$ of a particle in the rotating frame O',

$$\begin{aligned}
\vec{v} &= \frac{d\vec{r}}{dt}\Big|_O \\
&= \frac{d\vec{r}}{dt}\Big|_{O'} + \vec{\omega} \times \vec{r} \\
&= \vec{v'} + \vec{\omega} \times \vec{r}.
\end{aligned}$$

And the acceleration of the particle in the rotating frame O' is

$$\begin{aligned}
\vec{a} &= \frac{d\vec{v}}{dt}\Big|_O \\
&= \frac{d}{dt}\Big|_O \left(\vec{v'} + \vec{\omega} \times \vec{r} \right) \\
&= \frac{d\vec{v'}}{dt}\Big|_O + \frac{d}{dt}\Big|_O (\vec{\omega} \times \vec{r}) \\
&= \frac{d\vec{v'}}{dt}\Big|_{O'} + \vec{\omega} \times \vec{v'} + \vec{\omega} \times \vec{v'} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \\
&= \vec{a'} + 2\vec{\omega} \times \vec{v'} + \vec{\omega} \times (\vec{\omega} \times \vec{r}).
\end{aligned}$$

In this expression,

$$\begin{cases}
2\vec{\omega} \times \vec{v'} & \text{is the Coriolis acceleration} \\
\vec{\omega} \times (\vec{\omega} \times \vec{r}) & \text{is the centrifugal acceleration}
\end{cases}$$

8.6 Inertial and Non-Inertial Frames

The neuton equation observed in a inertial frame is

$$m\vec{a} = \vec{f}.$$

In a non-inertial frame, the equation of motion is modified to include the fictitious forces,

$$m\vec{a'} = \vec{f} - 2m\vec{\omega} \times \vec{v'} - m\vec{\omega} \times (\vec{\omega} \times \vec{r}).$$

So the Coriolis forces in a non-inertial frame is

$$\vec{f}_{\text{Coriolis}} = -2m\vec{\omega} \times \vec{v'}.$$

And the centrifugal force in a non-inertial frame is

$$\vec{f}_{\text{centrifugal}} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r}).$$

We can calculate the effective gravitational acceleration in a non-inertial frame by combining the real gravitational acceleration $\vec{g}$ with the fictitious accelerations,

$$\begin{aligned} \vec{g}_{\text{eff}} &= \vec{g} - 2\vec{\omega} \times \vec{v'} - \vec{\omega} \times (\vec{\omega} \times \vec{r}) \\ &= g_0 - \omega^2 r \cos\theta. \end{aligned}$$